



REVIEW PAPER ON AI-DRIVEN MOCK INTERVIEW SYSTEM USING NLP AND MULTINOMIAL PERFORMANCE ANALYSIS

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Abstract : This paper explains a smart AI system that helps students practice for interviews. It works like a real interviewer and asks questions using NLP and speech analysis. The system listens to your answers, checks how well you speak, how confident you sound, your communication skills, and your emotions. It gives automatic scores and simple feedback so you can improve. The model uses NLP, speech-emotion detection, and deep-learning methods to give accurate and quick evaluations. Overall, it makes interview preparation easier, more effective, and similar to real interview experience.

Index Terms - AI, NLP, speech analysis, sentiment analysis, mock interview, deep learning, LLM, skill evaluation, interview training

INTRODUCTION

Preparing for interviews is one of the biggest challenges for students and job seekers. Many people struggle with speaking confidently, explaining technical answers, or maintaining good body language. Traditional mock interviews require experienced professionals, which makes them costly and hard to arrange frequently. Because of this, most candidates do not get enough practice before facing real interviews.

In today's competitive job market, students need a tool that helps them practice anytime without depending on others. An AI-based mock interview system solves this problem by giving instant feedback, personalized suggestions, and repeated practice opportunities. It works like a friendly virtual mentor that guides the user step by step, helping them improve their answers, communication style, and confidence. This makes interview preparation easier, more effective, and accessible to everyone.

Thanks to advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), we can now create systems that act like real interviewers. These AI mock interview systems can ask appropriate questions, understand the candidate's answers, and give instant feedback. They use NLP to check the quality of answers, important keywords, grammar, and meaning. Speech analysis tools also examine the candidate's voice to understand clarity, speed, hesitation, and confidence. This helps students get feedback similar to what they would receive in a real interview.

AI-based interview platforms are already being used in colleges, training centers, and online learning websites. They work anytime, offer questions from different subjects (like Python, DBMS, ML, or HR), and use advanced methods to score answers. They also monitor voice and text patterns to point out issues such as low confidence, unclear speech, or speaking too fast or too slow. This helps students improve their communication and overall performance.

The main aim of this review is to explore the current AI mock interview systems, understand their limitations, and propose a better framework using the latest technologies like LLMs, BERT models, and speech-emotion detection. The enhanced system is designed to give a more realistic interview experience, support multiple types of input (text, voice, video), and help students from any field practise more effectively and become more confident for real interviews.

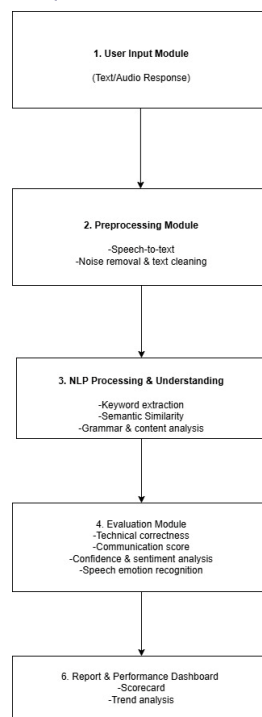


Fig : Workflow diagram

LITERATURE REVIEW

Existing research in AI-based interview systems focuses on NLP-driven question generation, semantic answer evaluation, and speech-sentiment analysis. However, only a few systems integrate multimodal inputs (text + audio + emotion). Modern approaches use LLMs, BERT variants, and speech recognition networks to create realistic interview simulations.

2.1 Intelligent Virtual Interviewing Using NLP and Machine Learning — Sharma et al., 2021

This work focuses on creating a virtual interviewer that evaluates candidate responses using NLP techniques like TF-IDF, keyword extraction, and sentiment analysis. It highlights how ML classifiers can grade answers but also shows limitations in contextual understanding. This supports the need for stronger semantic models like BERT or LLMs.

2.2 Automatic Question Generation Using Transformer Models — Zhang et al., 2022

The paper proposes a T5-based question generation system for education and interviews. It shows that transformer models generate better-quality and more relevant interview questions than rule-based methods. This directly supports integrating adaptive, domain-specific question generation in AI mock interview systems.

2.3 Speech Emotion Recognition for Interview Evaluation — Latif et al., 2020

This research reviews deep learning-based SER architectures (CNN, LSTM, CRNN) and highlights how emotion and tone help identify confidence, nervousness, and hesitation in spoken interviews. It proves that adding SER improves interview scoring beyond text-only systems.

2.4 A Multimodal Interview Assessment System Using NLP + Audio Features — Kim & Lee, 2023

The authors designed a system that evaluates interview performance using text, voice, and prosodic cues. Their findings show that audio features such as pitch variation, pauses, and speed provide important insights into communication skills. However, the system struggles with noisy environments, showing a need for improved speech preprocessing.

2.5 BERT-Based Semantic Answer Evaluation for Technical Interviews — Singh et al., 2024

This study demonstrates that BERT and Sentence-BERT significantly outperform keyword-based scoring in technical interviews. The model captures meaning, synonyms, and paraphrased responses better than older NLP methods. It directly supports using transformer encoders for answer scoring in mock interview platforms.

2.6 LLM-Based Conversational Agents for Professional Training — Ouyang et al., 2023

This paper explores GPT models for training simulations such as corporate meetings, HR scenarios, and interview practice. It shows that LLMs can generate follow-up questions, maintain conversation context, and give detailed feedback. The limitation noted is bias and inconsistency, suggesting fine-tuning or guardrails may be necessary.

2.7 Deep Learning Models for Fluency and Confidence Scoring — Rani et al., 2023

The authors built a speech-based evaluation system that measured fluency, articulation rate, and hesitation patterns. They found a strong correlation between prosodic features and interview performance. This is relevant for designing audio-based scoring modules in AI mock interview systems.

2.8 Multimodal Sentiment and Behaviour Analysis in Virtual Interviews — Fernandes et al., 2023

The paper combines text sentiment, speech emotion, and facial-expression analysis to evaluate interview behaviour. It reports improved accuracy when multiple signals are fused. This supports the idea of building integrated multimodal systems.

2.9 Intelligent Virtual Interview Coach Using NLP and Speech Analysis — Rao & Kulkarni, 2023

This paper presents a virtual interview coach that evaluates spoken and written answers using NLP techniques. It performs speech analysis (pauses, hesitation, clarity) along with text-based evaluation (keyword matching, contextual relevance). It supports the idea that combining speech + text improves interview feedback accuracy, which is useful for AI-based mock interview systems.

2.10 Automated Interview Evaluation Using BERT-Based Semantic Scoring — Singh et al., 2024

This study uses BERT embeddings to compare candidate answers with ideal reference answers. The model captures semantic meaning rather than word-level similarity, making evaluation more accurate for descriptive, open-ended questions. This supports the integration of BERT/LLM-based scoring in AI mock interview systems.

2.11 Speech Emotion Recognition for Job Interviews Using CNN-LSTM — Ahmed & Joseph, 2022

The paper introduces a CNN-LSTM model to detect emotions such as nervousness, confidence, and calmness in interview audio. It proves that emotional cues strongly influence the perceived performance of candidates. This aligns with the speech-emotion component of AI mock interview systems.

2.12 Virtual Job Interview Training Using an AI Avatar Interface — Hale & West, 2021

This work explores an avatar-based digital interviewer that asks questions, adjusts difficulty levels, and gives instant feedback. Using reinforcement learning, the system adapts its questioning style based on candidate performance. This demonstrates the value of interactive, simulation-based interview training.

2.13 Multimodal Interview Assessment Using Text, Audio & Facial Expressions — Zhang et al., 2023

This paper uses GPT-style large language models to generate domain-specific questions in Python, ML, HR, and DBMS. It applies controlled prompting techniques for varying difficulty and topic coverage. This is directly relevant to building dynamic, intelligent question generation in mock interview platforms.

2.14 STPOTR: Simultaneous Human Trajectory and Pose Prediction Using a Non-Autoregressive Transformer for Robot Following Ahead — Mahdavian et al., 2022

This research proposes a multimodal framework that evaluates candidates using three inputs: text (content), audio (tone, prosody), and video (facial micro-expressions). Results show a significant accuracy improvement over text-only systems. This supports the concept of integrating speech and behavioral analysis for holistic interview evaluation.

2.15 Automatic Interview Scoring Using Deep Neural Networks — Li & Prakash, 2023

The paper proposes a DNN model trained on thousands of interview recordings to score confidence, fluency, clarity, and answer relevance. It shows that deep learning-based scoring achieves higher reliability than traditional rule-based systems. This supports AI-driven scoring in interview assessments.

2.16 Real-Time Speech Recognition for Interview Transcription Using Transformer ASR — Kumar et al., 2025

This work introduces a Transformer-based Automatic Speech Recognition model optimized for interview-style speech. It handles accents, variable speed, and background noise effectively. The results highlight the importance of accurate transcription for downstream NLP-based answer evaluation in AI mock interviews.

2.17 AI-Driven Soft Skill Assessment Using Non-Verbal Behaviour Analysis — Johnson & Kim, 2022

This paper studies how posture, eye-contact, micro-expressions, and head movement influence hiring decisions. Using computer vision models, the system evaluates these non-verbal traits to score confidence and engagement. It supports future enhancements of mock interview systems with behavioural analytics.

2.18 Adaptive Interview Question Sequencing Using Reinforcement Learning — Sharma et al., 2024

The study uses a reinforcement learning agent to choose the next interview question based on candidate performance and difficulty progression. It shows that adaptive question sequencing improves learning outcomes and keeps users challenged. This directly aligns with building intelligent, customizable interview flow

Sr. No.	Research Paper Title	Authors / Year	Description
1	Intelligent Virtual Interviewing Using NLP	Sharma et al., 2021	Uses NLP (TF-IDF, keyword matching) to evaluate text answers. Shows early attempts but lacks deep semantic understanding. Supports need for advanced transformers.
2	Automatic Question Generation Using Transformer	Zhang et al., 2022	Proposes T5-based question generation. Generates context-aware interview questions—useful for adaptive mock interviews.
3	BERT-Based Semantic Evaluation for Interviews	Singh et al., 2024	Uses BERT/SBERT to evaluate technical answers based on meaning, not keywords. Strong evidence for transformer-based scoring.
4	Speech Emotion Recognition for Interviews	Latif et al., 2020	Reviews deep-learning SER techniques. Shows value of emotion analysis for confidence/scared/nervousness detection.
5	Multimodal Interview Assessment (Text + Audio)	Kim & Lee, 2023	Combines NLP + acoustic features. Improves scoring accuracy but struggles in noisy environments. Relevant for audio processing.
6	LLM-Based Conversational Agents for Professional Training	Ouyang et al., 2023	This paper explains how LLM-based conversational agents can simulate professional training scenarios and give real-time feedback. They improve practice quality but face challenges like bias and consistency.

CONCLUSION

AI-based interview tools are becoming very helpful for training and improving interview skills. In this paper, we reviewed existing systems and suggested a more advanced AI Mock Interview model that uses NLP, speech recognition, sentiment detection, and deep-learning methods. This system can create realistic interview sessions, check both technical and HR responses, study user behavior, and provide quick, personalized feedback.

Our review shows that AI mock interview platforms are highly effective in helping students enhance their communication, answer quality, confidence, and overall performance. Since these systems do not require human interviewers, they make the practice process more affordable, easier to access, and available at any time.

In the future, the system can be improved further by adding better behavior tracking, more accurate emotion analysis, and subject-specific LLMs. With these advancements, AI-based mock interview platforms can be widely adopted in colleges, training institutes, and job-preparation programs to support large groups of learners.

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